

Ordinary Least Squares Method:

$$\text{Residual} = y_i - \hat{y}_i$$

$$\begin{aligned}\therefore \text{Sum of square residual} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2\end{aligned}$$

$$\text{We can write, } f(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

We have to estimate b_0 & b_1 for which the function $f(b_0, b_1)$ should be minimized.

$$\text{So, } f(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 = 0$$

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) = 0 \quad \text{--- (i)}$$

$$\frac{\partial f(b_0, b_1)}{\partial b_1} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) x_i = 0 \quad \text{--- (ii)}$$

From equation (i),

$$\sum_{i=1}^n y_i - n b_0 - b_1 \sum_{i=1}^n x_i = 0$$

$$\Rightarrow n\bar{y} - n b_0 - b_1 \cdot n\bar{x} = 0$$

$$\Rightarrow n\bar{y} = n(b_0 + b_1 \bar{x})$$

$$\Rightarrow \bar{y} = b_0 + b_1 \bar{x}$$

$$\boxed{\therefore b_0 = \bar{y} - b_1 \bar{x}}$$

From equation (ii),

$$\sum x_i y_i - b_0 \sum x_i - b_1 \sum x_i^2 = 0$$

$$\Rightarrow \sum x_i y_i - (\bar{y} - b_1 \bar{x}) \cdot n\bar{x} - b_1 \sum x_i^2 = 0$$

$$\Rightarrow \sum x_i y_i - n\bar{x}\bar{y} + b_1 n\bar{x}^2 - b_1 \sum x_i^2 = 0$$

$$\Rightarrow \sum x_i y_i - n\bar{x}\bar{y} = b_1 (\sum x_i^2 - n\bar{x}^2)$$

$$\therefore b_1 = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$$